SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure **IP addressing** schemes, subnetting, and basic routing **techniques** using simulation tool, for efficient network design and topology. **(BL3**)

Assessment** Tool Weightage: **40%

QUESTIONS: 5

Annexure **A**

ANALYTICAL PROBLEM SOLVING

Total Marks:50

*Code of **Conduct:***

IS.V. Harish Kumar

(19251148 6)

(Name/Reg** No) certify **that this submission is my original** work **and

that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature *of the student:*

S.V. Yell

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has **decided** to **divide** this network into subnets for three different labs using **fixed subnet masks**. **Lab A** is assigned a /26 subnet, Lab **B** is assigned a /27 subnet, and Lab C **is assigned a /28** subnet. Based on **the** given subnet masks, apply subnetting concepts **to determine** the network address of each lab. Further, identify the valid range of host IP addresses and **the** broadcast address for each subnet. Finally, calculate **the total** number of **usable** host IP addresses **available** in each lab's subnet.

2. A network administrator **has** configured two systems with the following details: System 1 has **IP** address 192.168.1.130 **with** subnet **mask** 255.255.255.192, and System 2 has **IP** address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid **host IP** range for both subnets based on **the** given subnet mask. Calculate **whether each** IP address **falls** within its respective subnet range. Also, determine the number **of usable host addresses available** in each subnet.

3. An **organization has** been allocated the IP block **172.16.0.0/16** for its internal **network**. The network administrator **has** already decided the subnet masks for **each department** as follows: HR → /25, Sales → /26, IT →

/27, and Admin /**28**. Using the given subnet masks, apply subnetting techniques to assign **suitable subnet addresses** for **each** department. Determine the network address, **valid** host IP range, **and** broadcast address for each subnet. Based on the allocation, calculate **the** number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has **implemented subnetting** using a **/27 subnet mask** for its internal departments. Using **the given subnet** mask, apply subnetting techniques to calculate the total **number of subnets created** and the number of usable hosts per subnet. Determine **the subnet** to **which the IP** address **172.16.10.78** belongs. Identify the network address **and broadcast** address for that subnet. Also, determine whether the IP address **172.16.10.95 falls within** the same subnet range based on **the** given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into **its** shortest form and then expand the compressed address back to its full notation. Using a prefix length of /**64**, determine **the network** prefix **of the** given address. Also, calculate the total number of possible **host addresses available within this** network based on **the** given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; **misses key** elements	**Misinterprets** or **unable to understand problem**
Method **Selection**	20%	Selects most appropriate method/formu**la** with strong	Selects **correct** method with minor **errors**		

justification

Inappropriate or partially **correct** method

Logical, **systematic**

Solution **Process**

steps with

25**%**

correct

approach throughout

All

Mostly **correct** steps **with** minor

Disorganized **steps**; multiple **errors**

mistakes

calculations

Accuracy of Calculation

20%

accurate; **correct** final

Minor **calculation errors**

answer

Clearly **interprets**

Interpretation of **Results**

results with

15**%**

units and

engineering **relevance**

Unable to

select **method**

No clear

process

Frequent

errors; incorrect final

No valid calculations

answer

Basic interpretation with minor omissions

Limited or unclear interpretation

No

interpretation

3) Answer:-

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Course: Computer Networking (SA0704

Data :- 24/07/26

Subnetting of IP Block 172.16.0.0/16.

Given:-

An organization has been allocated the IP block 172.16.0.0 for its internal network. The subnet masks assigned to each department are:

* HR => /25 * Sales > /26
* IT - /27
* Admin -/28

The objective is to determine the network address, valed host range, broadcast, address, usable hosts， and the remaining unsised IP address range.

Step 1 : - Calculate the Number of Usable Hosts

Formul
a

=

* Total IP Addresses = 2^ (32 - Prefix Length)

&

Usable Hosts

Total IP Addresses -2

(The network address and bowoadcast address cannot be assigned to hosts).

Subnet
Allocation ! .

1) HR Department (125)

* Subnet Mask: 255.255.255.128
* Network Address: 172.16.0.0

A

• Valid Host Range: 172.16.0.1 -172-16-0.126

* Broadcast Address : 172.16.0.127 .

• Total IP Address : **128**

* Usable Hosts: 126

2) Sales Department

(126)

* Subnet Mask:

255.255.255.192 * Network

Address: 172.16.0.128

a valid **Host Range** :

172-16-0.129-172.16.0.190

* Broadcast Address:

172.16.0-191

* Total IP Address: 64

• Usuable Hosts: 62

3) It Department:.

& Subnet Mask:

*

255.255.255.
224

* Network Address:

172.16.0.192

* Valid Host Range :

172.16.0.193 - 172.16.0.222

* Broadcast Address ! .
18.2.16.0.223
* Total IP Address!.
& table Hosts : *30*

32

4 Admin
Department (das).

* Subnet Mask!.
255.255.255.24 o Network
Address :- 172.16.0.224
* Valed Host Range: 172-16.0.235-
172.16.0.238 * Broadcast Address :
172.16.0.239

- Usable Hosts: 14

A
Total IP Address: 16

Remaining Address Calculation in

* Total addresses in 172.16.0.0/16 = 65,536

• Total allocated = 128 +64 +32 +16 = 240 addresses * Remaining addresses = = 65, 536 -240 = 65, 296 addresses...

Conclusion

1.

The network 172.16.0.0/16 is successfully sullnettred using Variable Length subnet Masks (vesm'). The HR, Sales, IT, and Admin department receive 126, 62, 30, and 14 usable host addresses, respectively. The remaining address space (172.16.0.240 - 172.16.255.255) Contains 65, 296 IP addresses, which can be used for furture network expansion.

Subnetting of 172.16.10.0/24

Using a/27 subnet Mask

Given!-

1A Company uses the IP address 172.16.10.0/24 and has

implemented subnetting using internal departments.

Network Address: 172.16.10.0/24

a 127 subnet mask for th

Subnet Mask :-/27 (255.25

255.255.

5.224
)

The task is to:

:) Calculate the total number of subnets.

ii) Calculate the number of usable hosts per subnet: ppy Determine the subnet containing 172.16.10.78

iv) Find the **network** address and broadcast address of that subnet.

v) Check whether 172.16.10.95 belongs **to** the same subnet.

Step 1:- Calculate the Number of subnets:-

Original prefix

=124

New prefix

=/27

Number **of** borrowed bits =

27-24=3

Formula:-

Number of subnets = 23=8

Step 2:

Calculate the Number of usable Hosts:.

Hosts bits = 32-27=5

Formula:.

Total IP addresses per subnet = 25

= 32

Usable Hosts = 32~2 = 30

Step 3:- Determine the subnet of 172-16-10-78

The IP address 172.16.10.78 falls between:

172-16.10.65
172~16.10.94

* Network Address = 172.16.10.64

* Valid Host Range = 17.16.10.65-172-16.10.94 & Broadcast Address = 172-16.10.95

Final Answer:

× Original Network:-172-16-10-0/24

• Subnet Mask!. /27 (255.255.255.224) & Total

Number of subrets:.8

& usable Hosts per subnet:- 30